AKSHAY PRALHAD KANADE

Address: Prüfeninger Strasse 61, 93049 Regensburg, Germany
Mobile: (+49) 15215423232
E-Mail: akshaykanade77@gmail.com
LinkedIn: www.linkedin.com/in/akshay-kanade-746398122

PROFESSIONAL EXPERIENCE

Software Developer (AI/ML Engineer) - OneVision Software AG, Regensbur *June 2024 – Nov 2025*

- Designed end-to-end perception pipelines from ingestion to deployment, shortening delivery cycles by 50%.
- Boosted inference throughput by 70% through TensorRT tuning, ONNX export, and mixed precision execution.
- Benchmarked CNN, ViT, and foundation models, raising model selection efficiency by 40% across industrial imaging tasks.
- Introduced explainability and validation layers, reducing model related support queries by 30% and strengthening stakeholder trust.
- Coordinated with platform, DevOps, and product teams to operationalize models via CI/CD, enabling weekly production rollouts.

Working Student & Master Thesis (ADAS Validation, Computer Vision & ML Engineering) - AVL Software and Functions GmbH, Regensburg *Oct 2022 - Dec 2023*

- Built containerized object detection pipelines for ARM platforms, cutting inference latency by 50%.
- Reduced model footprint by 40-60% using pruning and quantization, enabling real time edge deployment.
- Automated ROS bag processing and annotation flows with Python/SQL, saving 30+ hours per month in manual effort.
- Verified LiDAR and radar outputs to ensure perception accuracy for ADAS validation across diverse test scenarios.

Data analyst - Accenture, Mumbai, India *Jan 2019 - Aug 2021*

- Created ETL workflows and automated reporting systems, improving data processing efficiency by 65%.
- Constructed SQL driven data pipelines supporting analytics and ML initiatives for enterprise clients.
- Partnered with cross functional teams to translate raw data into actionable insights for large scale operations.

EDUCATION

Hochschule Ravensburg Weingarten *Sep 2021- Oct 2023*
Master of Engineering: Electrical Engineering and Embedded Systems

Savitribai Phule Pune University *May 2014 – June 2018*
Bachelor of Engineering: Electronics and Telecommunication

SKILLS

- **Programming Languages:** Python, C++, SQL, C
- **Machine Learning & AI:** Deep Learning, Transfer Learning, Supervised & Unsupervised Learning, CNNs, Transformers, Vision Transformers (ViTs), Model Fine-Tuning
- **Computer Vision:** OpenCV, Detectron2, YOLO, Faster R-CNN, U-Net, ONNX, TensorRT
- **GenAI & LLMs:** Hugging Face Transformers, RAG, Prompt Engineering, Embeddings, FAISS
- **Frameworks & Libraries:** TensorFlow, PyTorch, Scikit-learn, Keras, Pandas, NumPy, Matplotlib, Seaborn
- **MLOps & Deployment:** MLflow, DVC, Docker, Kubernetes, Jenkins, CI/CD, FastAPI, REST APIs
- **Cloud Platforms:** Azure ML, AWS (S3, EC2, SageMaker), GCP

- **Optimization & Edge AI:** Pruning, Quantization, Mixed Precision, Edge Deployment
- **Data Annotation & Tooling:** CVAT, Label Studio, Dataset Curation, Data Augmentation, Quality Validation
- **Data Visualization & Analytics:** Power BI, Tableau
- **Systems & Tools:** Linux, Windows, Git, ROS, Jupyter, VS Code, PyCharm
- **Languages:** English (C1), German (A2, learning B1)

PROJECTS

- **AI Based Object Detection & Segmentation Pipeline**
  Implemented YOLOv5, Faster R-CNN, and U-Net models across 5+ datasets with >92% detection accuracy.
  Streamlined preprocessing pipelines using OpenCV and Albumentations, reducing preparation time by 40%.
  Optimized Jetson Nano inference using pruning and TensorRT, achieving 60% faster edge performance.

- **MLOps Deployment & Monitoring**
  Established reproducible workflows using MLflow, DVC, and Docker across more than 20 experiments.
  Implemented CI/CD pipelines, reducing model release time by 35%.
  Built monitoring dashboards tracking 10+ performance metrics and model drift.

- **LLM Based RAG Assistant**
  Developed RAG system using embeddings and FAISS, enabling sub second semantic search.
  Structured document chunking and prompts, improving response relevance by 30%.
  Deployed FastAPI based inference service supporting multi user concurrent requests.

- **Camera Calibration & Image Processing Automation**
  Performed intrinsic calibration using OpenCV, achieving a reprojection error of 0.0156.
  Applied edge detection, perspective correction, and geometric transformations to support automated vision workflows, reducing manual image adjustments by 50%.
  Validated calibration across multiple camera setups to ensure stable perception workflows.

CERTIFICATIONS

- Generative AI with Python and Hugging Face Transformers
- Computer Vision A–Z™: Learn OpenCV, SSD, YOLO, GANs and More
- AI Engineer Core Track: LLM Engineering, RAG, QLoRA, Agents
- Machine Learning, Data Science and Deep Learning with Python
- MLOps with MLflow, Docker, and Kubernetes for Deployment
- AI Engineer Production Track: Deploy LLMs & Agents at Scale
- AI Builder: Create Agents, Voice Agents & Automations in n8n
- AI Engineer Agentic Track: The Complete Agent & MCP Course
- DevOps Fundamental CI/CD with AWS + Docker + Ansible + Jenkins
- Python for Data Science and Machine Learning Bootcamp
- TensorFlow Developer Certificate

PUBLICATION

“Evaluation of dimensions of a vehicle using Velodyne and Blickfeld LiDAR”
International Journal of Scientific and Engineering Research (IJSER), Volume 12, Issue 12, Dec 2021

DRIVING LICENSE

Class **B**